

Indoor Microclimate Monitoring System

ESP-12E 5-parameter environmental monitor with custom analog dust channel

General Description

The MIC-5X continuously measures five microclimate parameters — temperature, humidity, barometric pressure, illuminance and airborne dust — with local OLED readout and remote access through a built-in Wi-Fi web server. Four parameters use commercial digital sensors; dust concentration is measured by a self-designed analog optical (nephelometric) channel built around a transimpedance amplifier. The system targets low-cost greenhouse and indoor air-quality monitoring.

Revision C summary. Wi-Fi core migrated from a NodeMCU module to a bare **ESP-12E** on a custom two-layer PCB with on-board bootstrapping, programming header and a full 5 V input-protection chain. The dust-channel output is re-scaled by an on-board resistive divider to the ESP-12E native ADC range (~ 1 V). Forced airflow is provided by a 2510 fan in the 3D-printed enclosure. All figures were re-derived for Rev. C; values requiring CFD, thermal or bench validation are marked as such rather than estimated.

Key Features

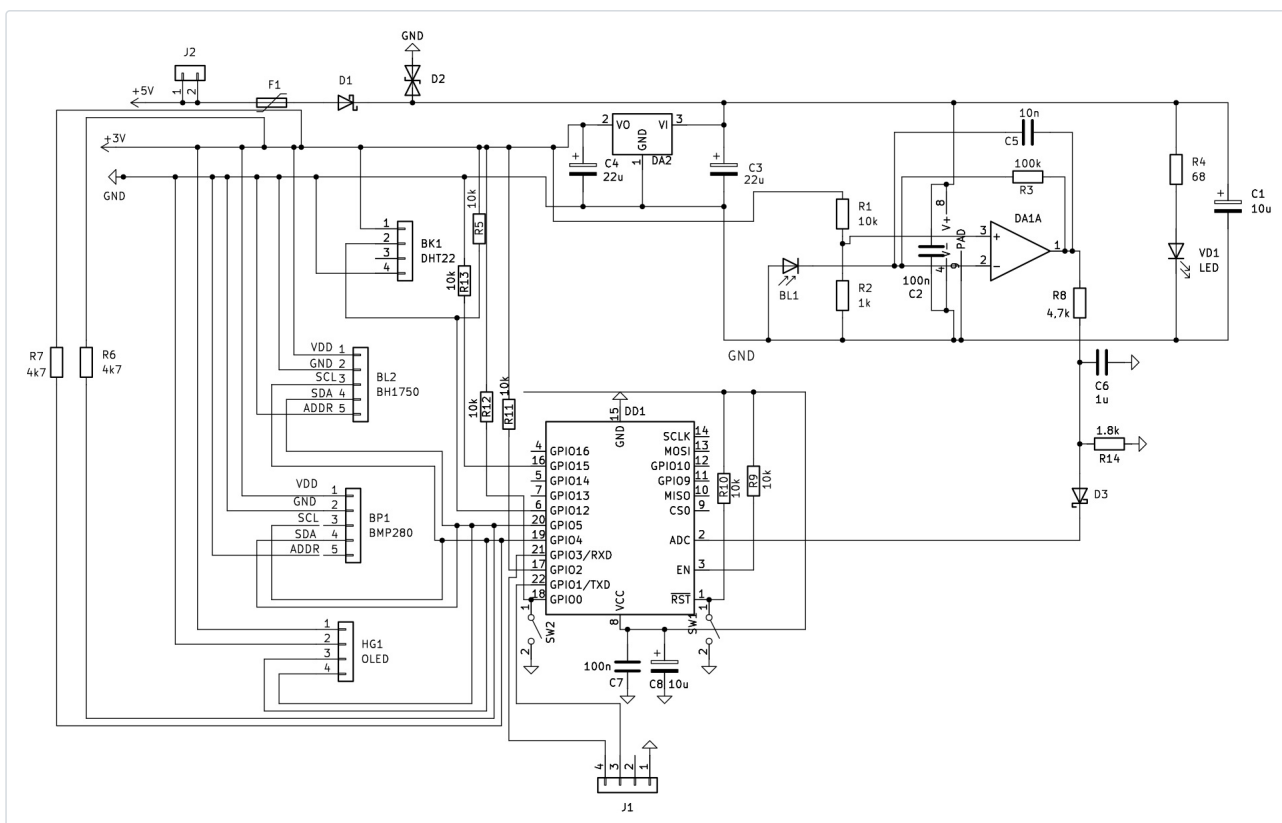
- 5 measured parameters
- Bare ESP-12E core, built-in Wi-Fi
- Local OLED + web server
- Custom analog dust channel
- Transimpedance front-end (LM358)
- 10-bit ADC scaled to ~ 1 V
- I²C digital sensor bus
- Single 5 V supply, 3.3 V LDO rail
- 5 V protection: PolyFuse + Schottky + TVS
- ADC over-voltage clamp
- Two-layer PCB (KiCad)
- 3D-printed enclosure, 2510 fan seat

01 System Overview

Block	Part	Function
MCU / Wi-Fi	ESP-12E (bare, DD1)	Acquisition, processing, Wi-Fi web server
Temp / humidity	DHT22 (BK1)	Temperature, relative humidity
Pressure	BMP280 (BP1)	Barometric pressure (+ backup temperature)
Illuminance	BH1750 (BL2)	Ambient light (I ² C)
Display	OLED (HG1)	Local readout (I ² C)
Dust channel	BL1 photodiode + VD1 IR LED + LM358 (DA1)	Analog $\rightarrow$ ADC, relative dust level
Regulator	AMS1117-3.3 (DA2)	5 V $\rightarrow$ 3.3 V
Input protection	PolyFuse F1, Schottky D1, TVS D2	Over-current / reverse / transient on 5 V
ADC clamp	Schottky D3 (SOT-23)	Over-voltage clamp on ADC node

Designators per Rev. C KiCad BOM and silkscreen.

1.1 Schematic



MCU boot-strapping & programming. The bare ESP-12E requires external strapping (vs the previous NodeMCU). The board provides the boot/flash strap and reset network from the 10 kΩ group (R1, R5, R9-R13), the two control buttons (SW1 reset, SW2 flash) and a 4-pin programming/UART header (J1).

02 Absolute Maximum & Key Ratings

Parameter	Condition	Value
Supply voltage (input)	External adapter / USB	5.0 V
Logic / sensor rail	AMS1117-3.3	3.3 V
ADC input range	ESP-12E TOUT	0–1.0 V
ADC resolution	10-bit nominal; ESP8266 TOUT ENOB ~8-9 bits in practice	1024 codes
PolyFuse hold current	F1 (PPTC, 1206); Wi-Fi TX peaks (~430 mA total, ms-scale) ride through — PPTC trip is thermal (~s-scale)	0.4 A
Supply current, average	ESP-12E modem-on ~70–80 mA + LED ~48 mA + sensors/OLED ~15 mA (estimate, not bench-measured)	~140 mA
Supply current, peak	Wi-Fi TX burst ~350 mA (ESP8266) + LED + sensors (estimate)	~430 mA
Sampling period	non-blocking (millis)	30 s
Dust LED current	VD1 via R4 = 68 Ω, after D1 drop (§3.1)	~48 mA

2.1 Recommended Operating Conditions

Parameter	Condition / note	Min	Typ	Max
Supply voltage U_{in}	USB / adapter, into protection chain; min set by D1 drop (~0.4 V) + AMS1117 dropout (~1.1 V at Wi-Fi TX peaks)	4.75	5.0	5.5
Ambient temperature	limited by LM358 (0...+70 °C standard grade) and FDM enclosure (PLA softens ~55–60 °C); digital sensors themselves are rated wider	0	25	+50
Relative humidity	non-condensing (DHT22)	0	—	100 %
ADC input at TOUT	after R8/R14 divider	0	—	1.0 V

Units in volts / °C unless noted. The temperature range is set by the **weakest links** — the standard-grade LM358 (0...+70 °C) and the printed PLA enclosure — not by the digital sensors (DHT22/BMP280 are rated to -40 °C). A wider range requires LM358B/LM2904 (-40...+85 °C) and PETG/ASA. Below ~4.75 V input the 3.3 V rail may lose regulation during Wi-Fi TX bursts (D1 drop + AMS1117 dropout); a low- V_F Schottky or P-MOSFET reverse protection and an LDO with lower dropout would recover the 4.5 V floor.

2.2 Pin / Connector Configuration

Ref	Connector / net	Pins	Function
J1	Programming header	4	UART TX/RX + GND/3V3 for ESP-12E flashing
J2	Power input	2	External 5 V into protection chain (F1→D1→D2)
SW1 / SW2	RESET / FLASH	—	Boot-mode strapping for programming
BK1	DHT22 header	3	Single-wire (proprietary, not Dallas 1-Wire) temperature / humidity
BL2	BH1750 header	4	I ² C illuminance
BP1	BMP280 header	4	I ² C pressure
HG1	OLED header	4	I ² C SSD1306 display
BL1 / VD1	Photodiode / IR LED	—	Optical pair, analog dust channel

Designators per Rev. C KICad schematic and BOM. Fan (2510) is wired to an external 5 V source, not to the board.

03 Analog Dust Channel

The dust channel is the only fully custom block. An IR LED illuminates the measurement zone; dust particles scatter light; a photodiode collects the scattered light; the photocurrent is converted to voltage by a transimpedance amplifier (TIA), then filtered and scaled to the ADC.

3.1 IR Emitter Drive

LED current set by series resistor R4 = 68 Ω:

$$I_{LED} = (5 - V_{D1} - V_F) / R4 = (5 - 0.4 - 1.35) / 68 \approx 48 \text{ mA} \quad (\sim 48 \% \text{ of a } 100 \text{ mA device limit})$$

Running the LED well below its rating favors long-term optical stability. The reverse-protection Schottky D1 (~0.4 V) is in series with the 5 V rail, so the effective LED supply is ~4.6 V; the earlier ~54 mA figure (which ignored D1) is superseded.

Continuous-drive caveats. VD1 is driven DC (no GPIO switching or modulation). Two consequences: (1) ~48 mA of continuous dissipation near the optical chamber adds a local thermal gradient on top of the ambient drift budgeted in §4; (2) the channel measures the DC scattered-light level, so any ambient IR entering the open-top chamber adds directly to the signal. A pulsed-LED / synchronous (LED-on minus LED-off) scheme would reject both and is listed in Future Work (§8).

3.2 Transimpedance Amplifier

Photocurrent converted by an LM358 (DA1) TIA, feedback R3 = 100 kΩ:

$$\text{Gain} = R3 = 100 \text{ k}\Omega \quad (100 \text{ mV per } \mu\text{A of photocurrent})$$

Raw TIA output spans ~0.3–3.3 V over the simulated 0–30 μA range, before the output network re-scales it to the ADC. *The LM358 input bias current (I_b ~45 nA) across R3 = 100 kΩ adds a ~4.5 mV offset term, small against the ~0.84 V working span but temperature-dependent; it is included in the error budget (§4). The LM358 is adequate here because the dust signal is slow (< 100 Hz) and its ~1.1 MHz GBW leaves ample loop gain; its poor behaviour near the rails is avoided by the reference offset (§3.3).*

3.3 Reference Offset

A divider from the clean 3.3 V rail lifts the operating point off zero, avoiding LM358 non-linearity near ground and setting the ~0.08 V floor on the transfer curve (§3.6).

3.4 Output Network — Stability Pole, LPF & ADC Scaling

The output path performs three jobs and is the most-revised part of Rev. C:

- **TIA stability pole — C5 = 10 nF across R3.** Compensates the photodiode-capacitance pole and stabilizes loop phase:

$$f_p = 1 / (2\pi \cdot R3 \cdot C5) = 1 / (2\pi \cdot 100 \text{ k}\Omega \cdot 10 \text{ nF}) \approx 159 \text{ Hz}$$

- **Output RC + divider — R8 = 4.7 kΩ, C6 = 1 μF, R14 = 1.8 kΩ.** R8/R14 scale the output to the ADC; R8||R14 with C6 sets the second corner:

$$\begin{aligned} \text{ratio} &= R14 / (R8 + R14) = 1.8 / (4.7 + 1.8) \approx 0.28 \\ f_{c2} &= 1 / (2\pi \cdot (R8 || R14) \cdot C6) = 1 / (2\pi \cdot 1.3 \text{ k}\Omega \cdot 1 \mu\text{F}) \approx 122 \text{ Hz} \end{aligned}$$

- **ADC clamp — Schottky D3.** Secondary protection only: the steady-state maximum at TOUT is already bounded by the divider itself (3.3 V rail-limited TIA × 0.28 ≈ 0.92 V < 1 V). D3 catches fault transients (e.g. C6 charge injection, probe slips). Note the ESP8266 TOUT absolute maximum is ~1.8 V per Espressif, so the divider — not the diode — is the primary guarantee.

Why this bandwidth. Airborne dust is a slow signal (reading changes over hundreds of milliseconds to seconds), so HF content is deliberately rejected. The two cascaded poles (~159 Hz, ~122 Hz) place the overall ~3 dB near **~90 Hz** — ample margin over the signal while suppressing out-of-band noise. A tighter band (C6 = 4.7 μF → ~26 Hz) is available with no board change (Future Work).

Target vs measured corner (honest note). An earlier first-order target of ~34 Hz used R8·C6 alone. Because R14 loads that node, the effective corner is set by R8||R14 ≈ 1.3 kΩ → ~122 Hz. Simulation and step response below confirm ~90 Hz (−3 dB), not 34 Hz.

3.5 ADC Scaling

The R8/R14 divider (~0.28) maps the raw TIA output (0.3–3.3 V) onto the ESP-12E ADC range (~1 V). Simulated post-divider signal spans **~0.08–0.92 V** over 0–28 μA — inside the 1 V window with headroom, clamped by D3.

3.6 Simulation Results (LTspice)

The plots below were redrawn from the LTspice simulation of the post-divider ADC node and are **simulated, not bench-measured**.

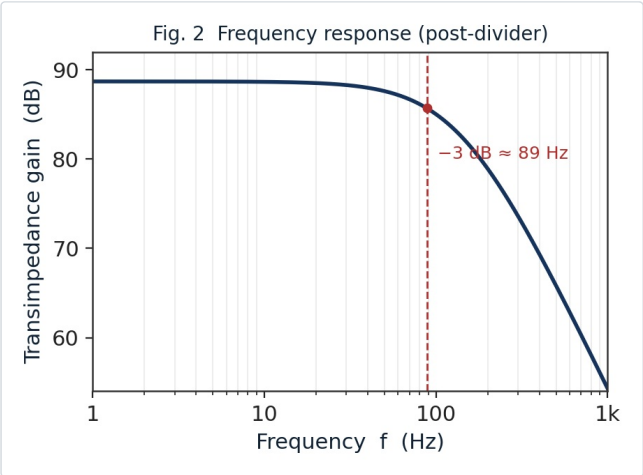


Fig. 2 — Frequency response. Flat ~ 88.7 dB; overall -3 dB at ~ 90 Hz from the cascaded ~ 159 Hz (R3·C5) and ~ 122 Hz (R8||R14·C6) poles.

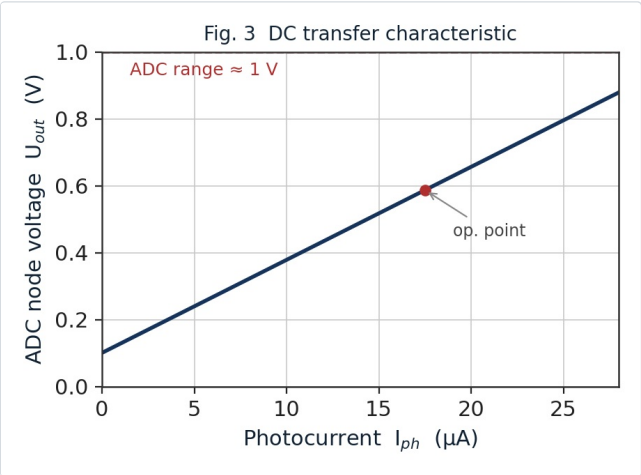


Fig. 3 — DC transfer $V_{out}(I_{photo})$. Linear, ~ 27.8 mV/ μ A after the 0.28 divider; ~ 0.1 V offset; span inside the ~ 1 V ADC range.

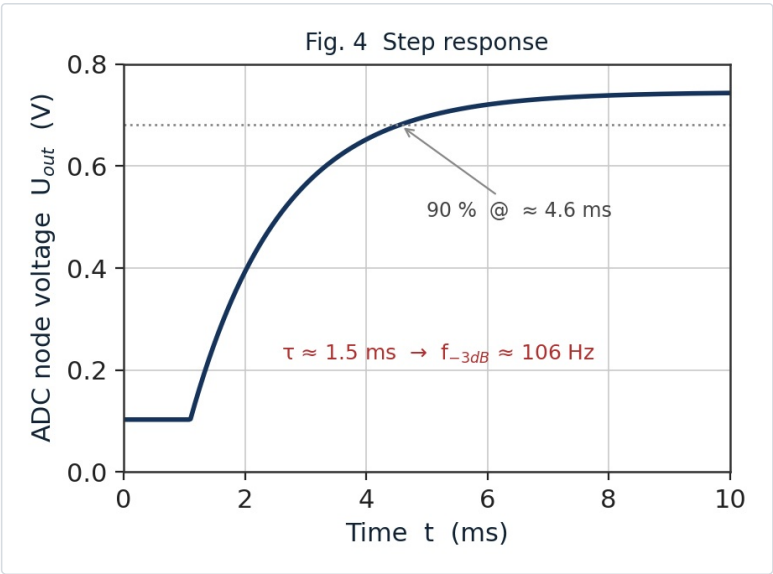


Fig. 4 — Step response. Near-first-order settling dominated by the lower pole; $\tau \approx 1.5$ ms (90 % by ~ 4.6 ms, slower than the single-pole 2.3τ because of the second cascaded pole). Equivalent single-pole $f_{-3dB} \approx 106$ Hz — consistent with the ~ 90 Hz two-pole corner of Fig. 2.

04Error Budget

Contributions over a 30 K ambient swing (worst case, before firmware compensation), combined by RSS. Values are engineering estimates from device datasheets, labelled as such — not bench-measured.

Source	Basis	Contribution
IR LED output drift	$\approx -0.4\text{ \%}/\text{K}$ optical power $\times 30\text{ K}$ (typically dominant)	$\sim 4\text{--}6\text{ \%}$
Photodiode responsivity drift	$\sim 0.1\text{ \%}/\text{K} \times 30\text{ K}$	$\sim 3.0\text{ \%}$
Reference / supply (V_{ref})	AMS1117 output tolerance & drift on the 3.3 V offset rail	$\sim 1.5\text{ \%}$
TIA feedback R3	100 k Ω , ± 100 ppm/K $\times 30\text{ K}$	$\sim 0.3\text{ \%}$
LM358 input bias current	$I_b \sim 45\text{ nA} \times R3 = \sim 4.5\text{ mV}$ vs $\sim 0.84\text{ V}$ span; drifts with T	$\sim 0.5\text{ \%}$
ADC quantization	1 LSB over $\sim 0.84\text{ V}$ span (10-bit)	$\sim 0.1\text{ \%}$
Total (RSS)	Dominated by optical (LED + photodiode) thermal drift	$\sim 5\text{--}7\text{ \%}$

The budget is dominated by the **optical** terms — LED output drift and photodiode responsivity drift — not by the electronics. Both track ambient temperature, so the BMP280 temperature reading is used for a first-order software correction (§5.2); the residual after compensation is the realistic accuracy figure. Absolute concentration additionally depends on calibration against a reference instrument (§8), which is out of scope for this revision.

05Improvements (Rev. B $\rightarrow$ C)

Derived analytically and verified in simulation, addressing the original weaknesses: excess bandwidth, thermal drift, absent supply protection, NodeMCU dependency.

5.1 Output Low-Pass Filter

The R3·C5 stability pole and R8||R14·C6 output filter match channel bandwidth to the slow dust signal and suppress out-of-band noise. Simulated overall corner ~90 Hz -3 dB (§3.4; Figs. 2 & 4). RMS noise ∝ √bandwidth, so this materially lowers output noise versus the original wide-band front-end.

5.2 Firmware Temperature Compensation

The temperature-dependent optical drift (LED + photodiode, the dominant terms in §4) is corrected in software using the temperature already measured by BMP280 — no added components. A first-order linear correction is expected to remove the bulk of the ~5-7 % swing; the residual (estimated low single-digit %) is set by the non-linear part of the drift and by calibration, and would need bench validation to state precisely.

5.3 Supply & ADC Protection

Element	Part	Function
Over-current	PolyFuse F1 (0.4 A hold)	Resettable over-current / short-circuit
Reverse polarity	Schottky D1 (SMA)	Reverse-polarity protection on 5 V
Transient clamp	TVS D2 (SMB)	Over-voltage / transient clamp on 5 V
ADC clamp	Schottky D3 (SOT-23)	Over-voltage clamp on ADC node

5.4 Forced Air Intake — 2510 Fan

A mechanical **2510 fan** seated in the enclosure wall provides forced cross-flow through the open-top chamber: inlet (fan side) → measurement zone → outlet on the opposite wall, sweeping dust through the illuminated zone at ~7 mm height so particles do not settle on the PCB. A solid central baffle between the optical pair blocks the direct LED→photodiode path — a defined, repeatable airflow rather than a thermal gradient.

Fan parameter (manufacturer-rated)	Value
Type / size	2510 brushless, 25×25×10 mm
Rated voltage	5 V DC
Current / power	0.1 A / 0.5 W
Speed	~7000 RPM
Airflow	~1.35 CFM (~0.64 L/s)
Static pressure	~2.32 mmH ₂ O
Noise	~18.6 dBA
Connector	2-pin XH2.54

Honest scope. The fan is **powered from a separate external 5 V source and is not electrically connected to the PCB** — not in the board BOM, no on-board drive or speed control. The enclosure provides only the mechanical seat. Actual in-chamber flow, air velocity and resulting response time are **not measured**; the rated airflow is a free-air figure. Quantifying in-chamber flow requires CFD or bench measurement (Future Work). Additionally, Rev. C drawings lack an exhaust port and a cover relief over the fan — see the enclosure errata in §6.6.

06Layout & Mechanical

6.1 PCB

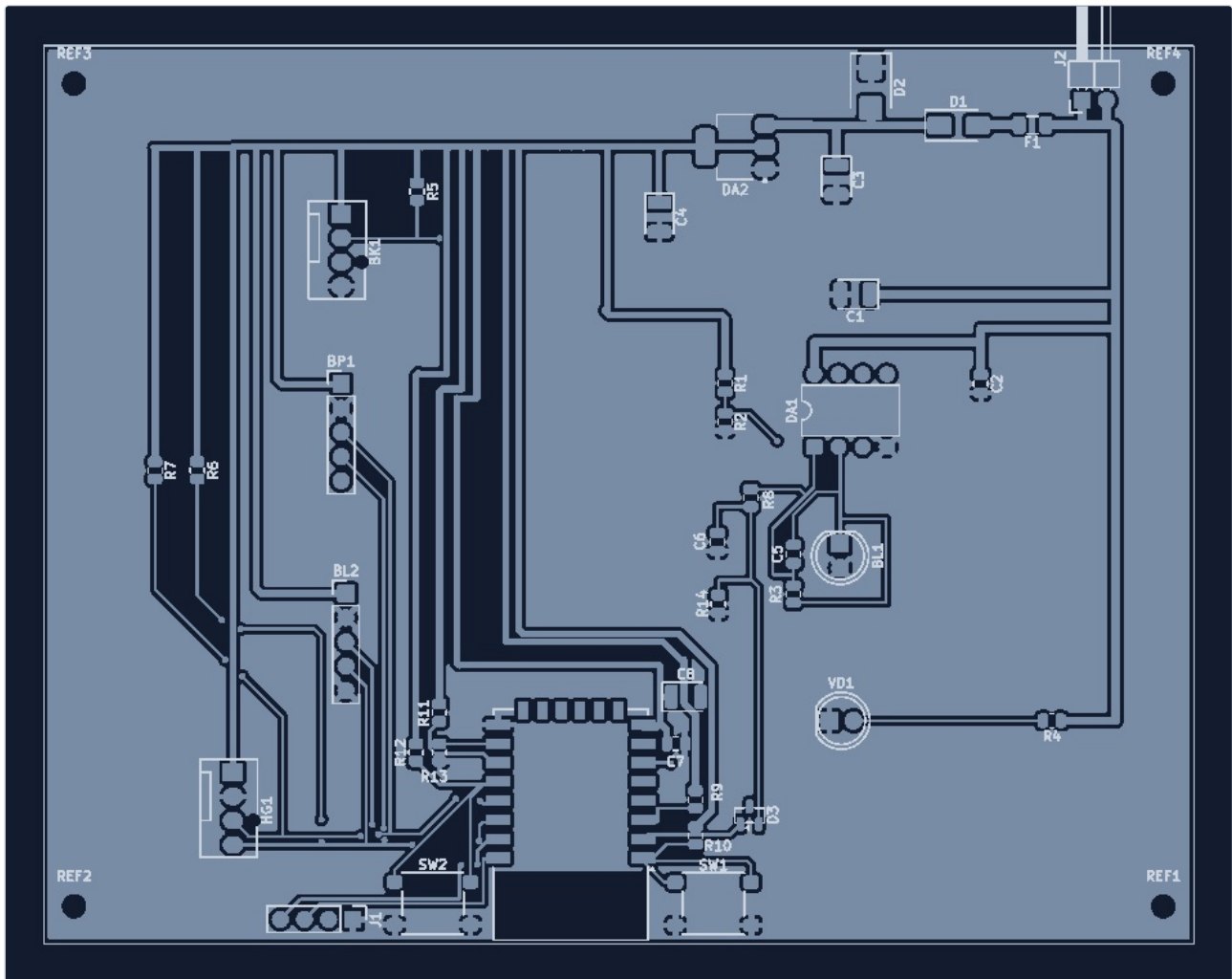
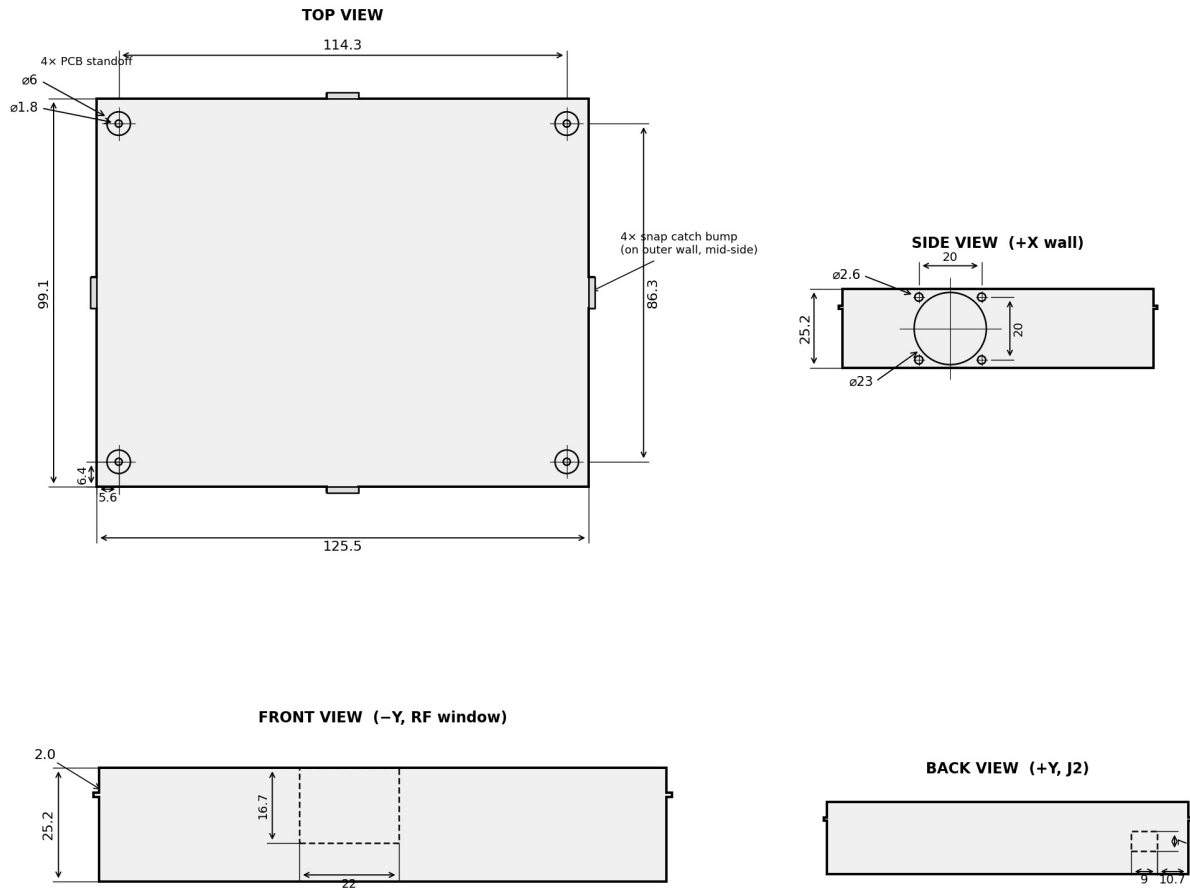


Fig. 5 — Two-layer PCB (top copper, silkscreen, board outline). Supply-protection cluster (F1/D1/D2/DA2) at the input edge; LM358 dust front-end and optical pair (BL1/VD1) grouped on the right; ESP-12E (DD1) central with strap network and programming header.

6.2 Enclosure — Part Drawings

MIC-5X · BASE — Dimensions (mm)

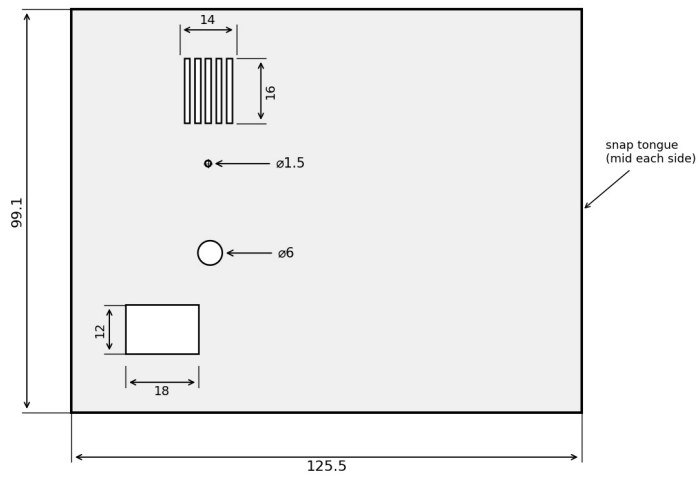


PCB on four ø6 standoffs (ø1.8 screw from below). Cover retained by 4 snap bumps on the outer walls (see Assembly). Corner radius R3. Fan 2510 external.

Fig. 6 — Base: three orthographic views (top, front, side) with dimensions in mm. Shows mounting-hole pitch, standoffs, wall/floor thickness and the ø23 fan bore on the +X wall.

MIC-5X · COVER — Dimensions (mm)

TOP VIEW (sensor vents)



FRONT VIEW



SIDE VIEW

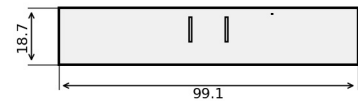
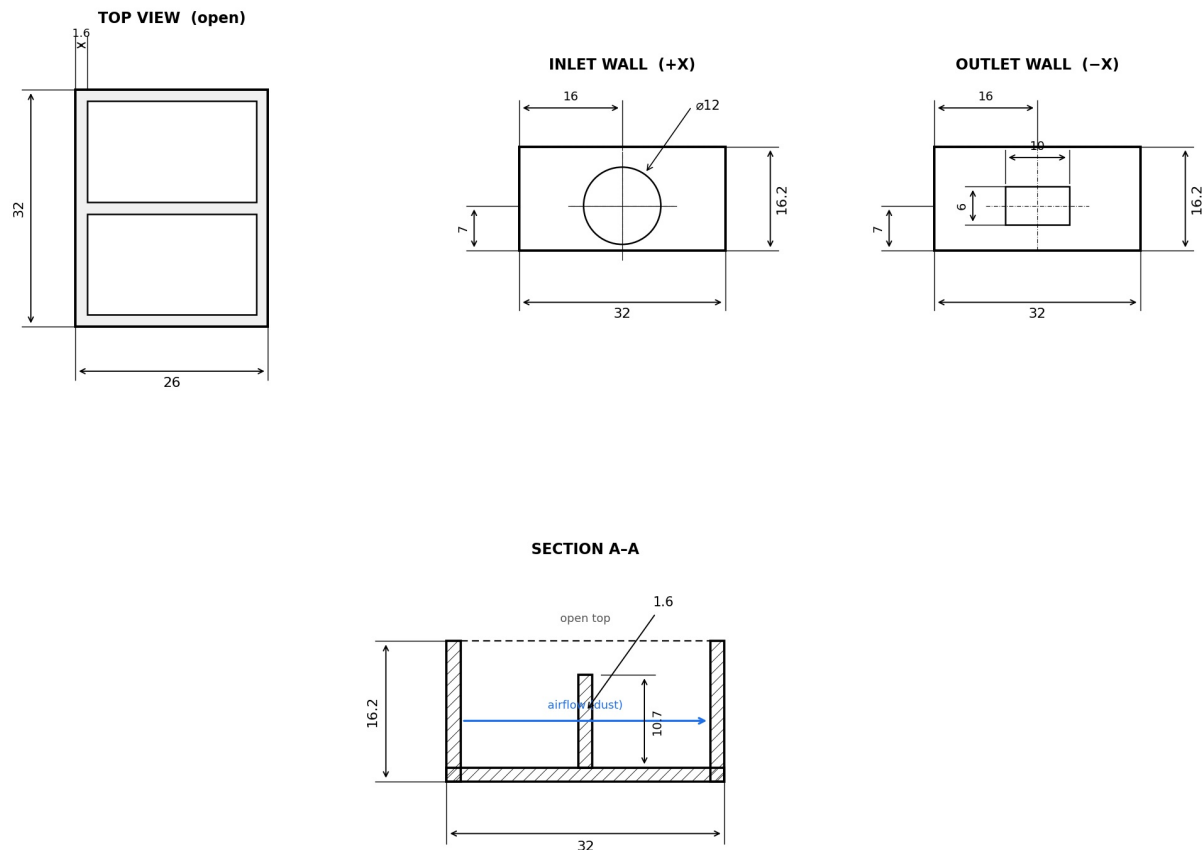


Fig. 7 — Cover: three orthographic views with dimensions in mm. Shows the sensor vents (OLED window, BH1750 $\varnothing 6$, BMP280 $\varnothing 1.5$, DHT22 grille) and the skirt that overlaps the base wall.

MIC-5X · OPTICAL CHAMBER (open-top) — Dimensions (mm)



Open-top cavity. Fan (blowing in) drives dust-laden air across the zone at ~7 mm height (inlet $\phi 12 \rightarrow$ zone $\rightarrow$ outlet 10×6). Solid baffle (1.6×10.7) blocks the direct LED $\rightarrow$ PD path; PD sees only scattered light. Walls 1.6.

Fig. 8 — Optical chamber (open-top): wall views with dimensions in mm and a cross-section A-A. The fan drives dust-laden air across the measurement zone ($\phi 12$ inlet $\rightarrow$ zone $\rightarrow 10 \times 6$ outlet at ~7 mm height), sweeping particles through so they do not settle on the PCB. The solid central baffle (1.6×10.7) blocks the direct LED $\rightarrow$ photodiode path; the photodiode receives only light scattered by dust.

Parametric CadQuery model (v9); STEP/STL exported for base, cover, chamber and assembly. Hidden contours shown dashed.

6.3 Assembly & Service Access

The enclosure is a three-part stack. The PCB is screwed down; the cover is retained by snap-fit tongues (no cover screws, no threaded plastic):

Step	Action
1 — PCB	The board rests on four $\phi 6$ standoffs (top at 5.0 mm) and is fixed from beneath the base floor by four ~2.2 mm thread-forming screws driven into $\phi 1.8$ pilot holes in the bosses (hole pitch 114.3×86.3 mm). The $\phi 1.8$ figure on Fig. 6 is the <i>pilot</i> diameter, not a screw size. If M3 hardware is substituted, the pilots must be opened to $\phi 2.5$ and the bosses grown to $\geq \phi 7$, otherwise a $\phi 6$ boss will split. Screw heads sit proud of the outer floor — recess them (counterbores) or fit ≥ 2 mm adhesive feet so the unit sits flat.
2 — Chamber	The open-top optical chamber drops over the emitter/detector pair and lifts off independently.
3 — Cover (snap-fit)	The cover skirt slides over the base wall. Four flexible tongues on the skirt (each framed by two slots) flex outward over small catch bumps on the base outer wall (~1.2 mm engagement, at ~19 mm height) and snap under them, retaining the cover without screws.

The snap-fit holds the cover against fan vibration while allowing tool-free access: press the tongues outward at the slots and lift the cover straight up; the PCB stays seated on its standoffs, giving full access to the top side (sensors, ESP-12E, J1 programming header). The optical chamber lifts out separately. Mating plane is at 25.2 mm; closed overall height 27.2 mm.

6.4 Sensor Mounting

The digital sensor modules (DHT22, BMP280, BH1750) and the OLED are pre-assembled breakout boards. Each plugs vertically into its 0.1" pin header on the PCB (BK1, BP1, BL2, HG1) and stands upright inside the enclosure. The cover has a matching vent or window directly above each one — a slotted grille over the DHT22, a $\phi 6$ port over the BH1750, a $\phi 1.5$ port over the BMP280, and an 18×12 window over the OLED — so the sensors sample ambient air (and the display is visible) through the closed cover. The optical dust pair (VD1 LED, BL1 photodiode) is soldered flat on the board and sits under the separate optical chamber.

6.5 Dust Airflow Path

The 2510 fan is seated in the base +X wall ($\phi 23$ bore) and **blows inward**. Its open area overlaps the chamber inlet ($\phi 12$) on the same side: with the bore centred mid-wall the two centre-lines differ by ~1–3 mm vertically, which is inside the ± 5.5 mm margin given by $\phi 23 \gg \phi 12$ (see §6.6, item C). The fan pushes outside air directly into the measurement zone; dust-laden air then crosses the zone (past

the illuminated volume, above the diodes) and leaves through the 10 × 6 outlet on the opposite chamber wall. This gives a defined, one-way sample flow rather than relying on diffusion.

Rev. C as drawn is incomplete here: there is no matching exhaust aperture in the −X base wall, so the chamber outlet discharges into the closed electronics cavity. The sample flow therefore recirculates dust over the PCB and sensors and pressurises the enclosure — the opposite of the design intent. See §6.6, item A for the required change.

6.6 Enclosure Errata & Required Changes (Rev. C.1)

Design review of Figs. 6–8 against the stated stack-up (floor 2.0 + standoff 5.0 + PCB 1.6 + in-chamber inlet height 7 ⇒ inlet centre ≈13.6 mm above the inner floor) found the following issues. They do not invalidate the concept, but must be fixed before the enclosure is printed as production Rev. C:

#	Issue	Consequence	Required change
A	No exhaust port. The chamber 10 × 6 outlet vents into the sealed electronics cavity; the only escape paths are the cover sensor vents.	Dust-laden air washes over the PCB, OLED and digital sensors and settles there; fan static pressure (rated ~2.32 mmH ₂ O ≈ 23 Pa ≈ 0.23 hPa) biases the BMP280 — comparable to the 0.3 hPa pressure-trend threshold (§7).	Add an aperture ≥ the chamber outlet (e.g. 12 × 8 mm) in the −X base wall, co-axial with the chamber outlet at ≈13.6 mm height; ideally duct the outlet through the wall.
B	Cover skirt / fan interference. The wall is 2.0 mm, the fan body 10 mm. Mounted on the outer face (M2.5 through the ø2.6 holes @ 20 mm pitch, nuts inside), the 25 × 25 fan frame occupies the wall region that the cover skirt (z ≈ 8.5...25.2 mm) slides over.	The cover cannot close with the fan fitted; Fig. 7 shows no relief.	Cut a skirt relief ≥26 mm wide over the fan footprint on the +X side (full skirt depth), or mount the fan inside — which then requires ≥10 mm clearance to the chamber inlet and a PCB keep-out. Self-tapping screws into 2 mm wall are not acceptable; use through-bolts with nuts.
C	Fan bore vs. inlet alignment. A ø23 bore truly co-axial with the inlet (centre ≈13.6 mm) would leave ≈0.1 mm of wall above the hole (wall top 25.2 mm) — i.e. it breaks the mating edge.	“Same height and centre-line” in Rev. C text was not geometrically achievable.	Centre the bore mid-wall (12.6 mm): the ~1 mm offset is harmless (ø23 » ø12, margin ±5.5 mm), and the top ligament grows to ≈1.1 mm — still thin for FDM: print the +X wall with ≥3 perimeters or reduce the bore to ø22.
D	Snap-fit strain in PLA. Catch bumps at ~19 mm engage tongues rooted at the cover top plate (27.2 mm), i.e. effective cantilever L ≈ 8.2 mm; with t = 2.0 mm and δ = 1.2 mm, ε ≈ 1.5·t·δ/L ² ≈ 5.4 % — far above PLA’s ~1.5–2.5 % limit.	Tongues whiten or crack on the first few open/close cycles.	Reduce engagement to ~0.6–0.8 mm and thin the tongue locally to ~1.2 mm (ε → ~1.6–2.1 %), add ≥45° lead-in chamfers; or print the cover in PETG/ASA.
E	Screw specification (§6.3, step 1).	“ø1.8 screws” was ambiguous; M3 into ø1.8 pilots splits the ø6 bosses; heads rock the unit.	~2.2 mm thread-forming screws into ø1.8 pilots (or M3 with ø2.5 pilots and ≥ø7 bosses); counterbores or ≥2 mm feet.
F	Stray-light and dust paths (minor). The chamber top (≈22.8 mm) sits 2.4 mm below the cover underside (25.2 mm); the OLED 18 × 12 window is an open aperture.	Ambient IR reaches BL1 via cover vents + the open chamber top (worsens the §8 ambient-light limitation); dust falls onto the display.	Matte-black cover underside above the chamber or a chamber lid with a labyrinth; optional thin acrylic glazing recessed in the OLED window.

07Firmware Summary

- **Non-blocking loop:** sensors polled every 30 s via `millis()` ; web server stays responsive (no `delay()`).
- **Sensor validation:** DHT22 retried up to 3×; out-of-range rejected; last valid value retained.
- **Dust averaging:** 16 ADC samples per reading, taken as a short burst (a few ms apart) and averaged. This reduces white noise (∝ 1/√N, ~×4) but not low-frequency drift, which is handled by the analog LPF (§3.4) and temperature compensation (§5.2). *Known ESP8266 quirk: TOUT readings taken during active Wi-Fi TX are disturbed; the sample burst should be scheduled away from network activity (or outliers rejected), otherwise occasional corrupted readings are expected.*
- **Temperature cross-check:** DHT22 and BMP280 redundancy.
- **Pressure trend:** 0.3 hPa threshold (rising / falling / stable).
- **Web server:** HTTP on port 80; page auto-refresh every 30 s.

08Limitations & Future Work

Item	Status	Planned improvement
Absolute dust calibration	Relative only	Two-point calibration vs reference instrument
Light→concentration model	Not quantified	Experimental Mie-scattering characterization
Optical-pair geometry	Empirical (17.3 mm)	Angle optimization from scattering diagram
Airflow through chamber	Forced (2510 fan); in-chamber flow not measured	CFD / bench flow characterization
Fan integration	External supply, not on PCB	On-board driver + PWM/tach (optional)
Channel bandwidth	~90 Hz −3 dB (simulated)	Tighten to ~26 Hz via C6 = 4.7 μF if noise-limited

Thermal drift	~5–7 % uncompensated (§4)	Firmware compensation via BMP280 (§5.2), residual low single-digit %
Ambient-light rejection	None — DC measurement, LED always on, open-top chamber admits ambient IR	Pulsed LED via GPIO + synchronous (on–off) sampling
3.3 V bulk capacitance	C3+C4 = 44 µF; marginal for ~350 mA Wi-Fi TX bursts	Add ≥220 µF bulk near ESP-12E to prevent brown-out resets
Operating temperature	0...+50 °C (LM358 grade, PLA enclosure)	LM358B/LM2904 + PETG/ASA for –20...+70 °C
Supply / ADC protection	Implemented	PolyFuse + reverse + TVS + ADC clamp (§5.3)
Chamber exhaust	Missing — outlet vents into the enclosure cavity, dust recirculates over PCB/sensors, fan pressure biases BMP280 (§6.6-A)	–X base-wall aperture (≥12 × 8) aligned with the chamber outlet
Cover-fan interference	Skirt collides with externally mounted 2510 (§6.6-B)	Skirt relief ≥26 mm over fan footprint; M2.5 through-bolts + nuts
Snap-fit material	~5.4 % tongue strain in PLA (§6.6-D)	0.6–0.8 mm engagement, thinner tongues, or PETG/ASA cover

All electrical figures re-derived for Rev. C and cross-checked against LTspice (Figs. 2–4). Items requiring CFD, thermal or bench validation are stated as such. Simulation results are labelled as simulated; no bench measurements are claimed.

09Revision History	
Rev.	Summary of change
B	Initial thesis design: NodeMCU module; dust ADC scaled 0.3–3.3 V via on-module 220k/100k divider; thermal-convection air intake; output LPF specified at ~34 Hz.
C	Migrated to bare ESP-12E on a custom two-layer PCB with boot strapping, RESET/FLASH buttons and a 4-pin programming header. Added full 5 V input protection (PolyFuse → Schottky → TVS) and ADC over-voltage clamp. Output re-scaled by on-board R8/R14 divider to the ~1 V ADC range; corrected effective corner to ~90 Hz –3 dB (two cascaded poles at ~159 and ~122 Hz). Replaced convection intake with a 2510 forced-air fan and an open-top optical chamber with cross-flow (inlet→zone→outlet) that sweeps dust through the measurement zone instead of letting it settle. Cover retained by four snap-fit tongues engaging catch bumps on the base wall (tool-free, no threaded-plastic screws). All figures re-derived; simulated values labelled as such.
C.1	Review errata: LED current corrected for D1 series drop (~54 → ~48 mA); operating temperature range restated as 0...+50 °C (limited by standard-grade LM358 and PLA enclosure, not by the sensors); minimum supply raised to 4.75 V (D1 drop + AMS1117 dropout at Wi-Fi TX peaks); D3 clamp role clarified (R8/R14 divider is the primary over-voltage guarantee); thermal-drift figure in §8 aligned with §4 (~5–7 %); added supply-current estimates, ambient-light / pulsed-LED limitation, 3.3 V bulk-capacitance note, and ESP8266 Wi-Fi-vs-ADC firmware caveat. Enclosure review (§6.6): missing –X exhaust port identified as a required change (dust recirculation + BMP280 pressure bias); cover-skirt relief over the external fan specified; fan-bore/inlet alignment corrected (mid-wall bore, ~1 mm offset accepted, thin top ligament flagged); snap-fit engagement reduced for PLA (strain ≈5.4 % → ≤2 %); PCB screw spec clarified (~2.2 mm thread-forming into ø1.8 pilots) with counterbore/feet note.

